



TRIOS CYBER

LABORATORY PRACTICE 27

MITRE ATT&CK Mapping Lab

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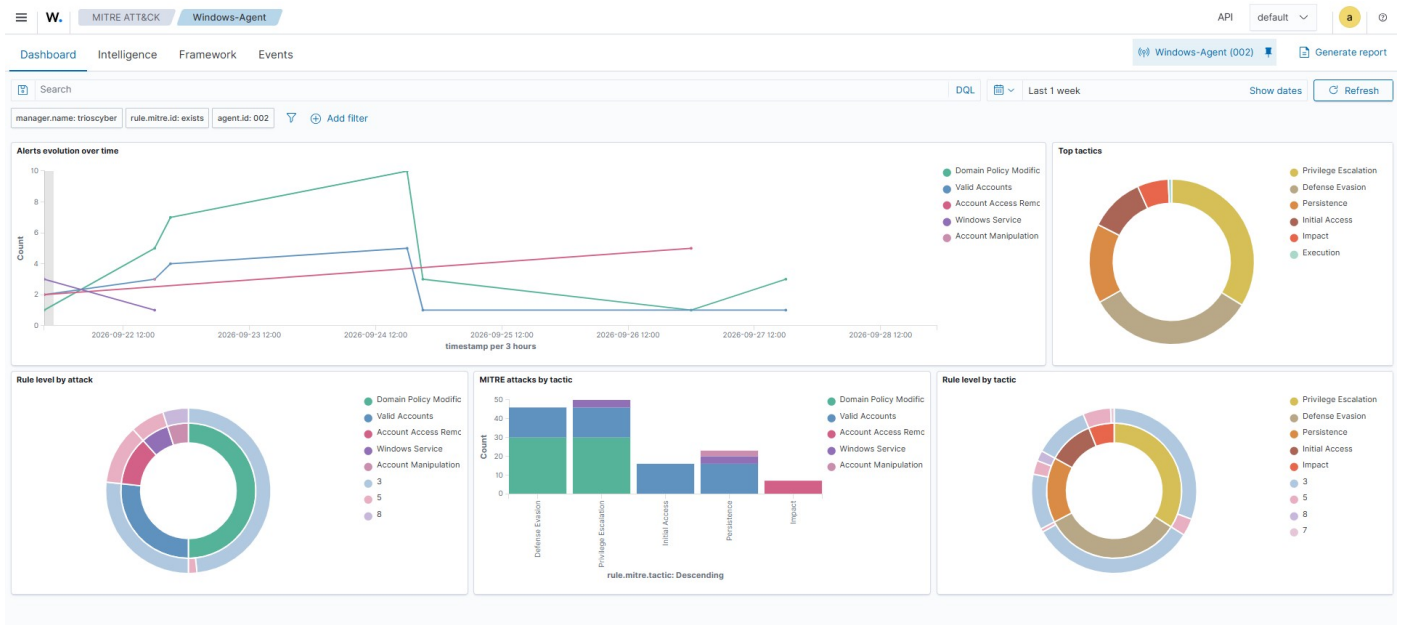
1. Laboratory Objective and Mapping Scope

This laboratory selected one simulated Windows behavior and mapped it to the most relevant MITRE ATT&CK technique using Wazuh ATT&CK telemetry and Windows event evidence. The selected behavior was the creation of a Windows service on the monitored endpoint. Supporting evidence was collected from the Wazuh MITRE ATT&CK view and the corresponding Windows EventChannel record.

Evidence item	Observed value	Purpose / result
Simulated behavior	Windows service creation	Behavior selected for ATT&CK mapping
Endpoint	Windows-Agent / ID 002 / 10.10.20.30	Monitored source endpoint
Windows event	System Event ID 7045	Service-installation evidence
Wazuh alert	Rule 61138 / Level 5	New Windows Service Created
MITRE ATT&CK	T1543.003 - Windows Service	Technique mapped by Wazuh

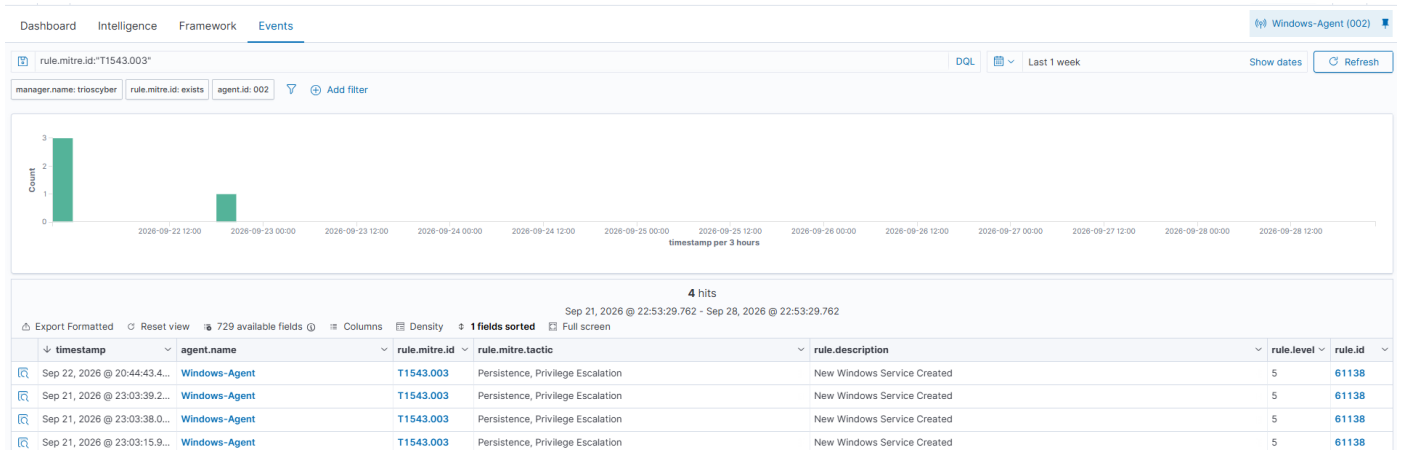
2. MITRE ATT&CK Activity Overview

The MITRE ATT&CK dashboard for Windows-Agent summarizes ATT&CK-mapped endpoint activity and shows Windows Service among the observed attack techniques. The dashboard also presents Persistence and Privilege Escalation among the active tactics, providing the broader ATT&CK context for the selected behavior.



3. Technique-Specific Filtering

The Events view was filtered with rule.mitre.id:"T1543.003". Four matching records were returned for Windows-Agent. Each record was classified as New Windows Service Created, rule 61138 at level 5, and mapped to the Persistence and Privilege Escalation tactics.



4. Simulated Behavior and Endpoint Evidence

The selected Wazuh document identifies the reporting endpoint as Windows-Agent (ID 002) at 10.10.20.30. The Windows System event records service TriosLab21, image path C:\Windows\System32\cmd.exe /c exit, LocalSystem as the service account, and demand start. Service Control Manager Event ID 7045 confirms that a Windows service was installed on the endpoint.

Field	Observed value	Interpretation
Agent	Windows-Agent / ID 002 / 10.10.20.30	Endpoint that generated the telemetry
Service name	TriosLab21	Simulated service behavior
Image path	C:\Windows\System32\cmd.exe /c exit	Service executable command
Service account	LocalSystem	Account recorded for the service
Start type	Demand start	Service configured for on-demand start
Windows event	System / Event ID 7045	Service installation recorded by Windows

Document Details

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Table JSON

t _index	wazuh-alerts-4.x-2026.09.23
t agent.id	002
t agent.ip	10.10.20.30
t agent.name	Windows-Agent
t data.win.eventdata.accountName	LocalSystem
t data.win.eventdata.imagePath	C:\\Windows\\System32\\cmd.exe /c exit
t data.win.eventdata.serviceName	TriosLab21
t data.win.eventdata.serviceType	servicio de modo usuario
t data.win.eventdata.startType	inicio por solicitud
t data.win.system.channel	System
t data.win.system.computer	DESKTOP-VOJCDSO
t data.win.system.eventID	7045
t data.win.system.eventRecordID	884
t data.win.system.eventSourceName	Service Control Manager
t data.win.system.keywords	0x8080000000000000
t data.win.system.level	4
t data.win.system.message	"Se instaló un servicio en el sistema. Nombre del servicio: TriosLab21 Nombre del archivo del servicio: C:\Windows\System32\cmd.exe /c exit Tipo de servicio: servicio de modo usuario Tipo de inicio de servicio: inicio por solicitud Cuenta de servicio: LocalSystem"
t data.win.system.opcode	0

5. Wazuh Rule and MITRE Mapping Evidence

The lower portion of the same Wazuh document preserves the correlation fields used for the ATT&CK mapping. Wazuh classified the event as New Windows Service Created under rule 61138 at level 5 and assigned MITRE ATT&CK T1543.003 - Windows Service. The mapped tactics are Persistence and Privilege Escalation.

data.win.system.opcode	0
data.win.system.processID	612
data.win.system.providerGuid	{555908d1-a6d7-4695-8e1e-26931d2012f4}
data.win.system.providerName	Service Control Manager
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-09-23T01:44:40.3165120Z
data.win.system.task	0
data.win.system.threadID	5704
data.win.system.version	0
decoder.name	windows_eventchannel
id	1790127883.184054
input.type	log
location	EventChannel
manager.name	trioscyber
rule.description	New Windows Service Created
# rule.firedtimes	1
rule.groups	windows, windows_system
rule.id	61138
# rule.level	5
rule.mail	false
rule.mitre.id	T1543.003
rule.mitre.tactic	Persistence, Privilege Escalation
rule.mitre.technique	Windows Service
timestamp	Sep 22, 2026 @ 20:44:43.428

6. MITRE ATT&CK Mapping Rationale

The mapping to T1543.003 - Windows Service is supported directly by the observed creation of service TriosLab21 and Windows Event ID 7045. Wazuh associates this technique with Persistence and Privilege Escalation. The LocalSystem service context, executable path, Service Control Manager provider, and service-creation alert provide the supporting evidence for that mapping.

Mapping result: T1543.003 - Windows Service | Persistence | Privilege Escalation

7. Validation Summary

Requirement	Evidence obtained	Status
Simulated behavior	TriosLab21 Windows service creation	Completed
ATT&CK mapping	T1543.003 - Windows Service; Persistence / Privilege Escalation	Mapped
Wazuh evidence	Rule 61138 / Level 5 / New Windows Service Created	Verified
Log evidence	System Event ID 7045; Service Control Manager	Verified
Supporting fields	Endpoint, image path, LocalSystem, start type, timestamp	Recorded

8. Conclusions

The selected Windows service-creation behavior was mapped successfully to MITRE ATT&CK T1543.003 - Windows Service. Wazuh preserved the corresponding rule, tactics, endpoint, service details, Windows Event ID 7045, and timestamp. Together, the Wazuh and EventChannel evidence directly support the Persistence and Privilege Escalation mapping required for the laboratory.